Enhanced Fabric Defect Detection With Feature Contrast Interference Suppression

Jinbao Wang, *Member, IEEE*, Jiayi Cheng, Can Gao, *Member, IEEE*, Jie Zhou, *Member, IEEE*, and Linlin Shen, *Senior Member, IEEE*

***Abstract*— Detecting fabric defects is a critical task in the textile industry, as it involves identifying and localizing imperfections in fabrics. However, many existing unsupervised defect detection methods are sensitive to variations and disturbances in the normal background of fabric images, leading to a high false alarm rate and unsatisfactory performance. To address this issue, we propose a novel defect detection method called feature contrast interference suppression (FCIS) that aims to enlarge the feature distance between foreground and background textures while suppressing the distribution of background features to obtain a more robust background representation. Our approach consists of several key components: First, we introduce background-foreground contrastive learning (BFCL) to enhance feature representation between background and foreground elements at the feature level. Second, we use multiscale denoising knowledge distillation to aid the model in learning more accurate representations of normal background features. Finally, we design an attention decoder that refines and segments the defect mask generated through feature comparison, resulting in improved detection accuracy. Our proposed FCIS has been validated using five mainstream datasets. Experimental results demonstrate that our proposed method outperforms existing approaches, achieving improvements of 2.38% in image-level AUROC and 3.73% in pixel-level AUPRO, along with a 9.12% enhancement in average defect localization precision.**

***Index Terms*— Attention segmentation, fabric defect detection, interference suppression, knowledge distillation, unsupervised contrastive learning.**

I. Introduction

FABRIC is an essential material in daily life, and fabric inspection plays a crucial role in ensuring product quality. Due to the variety of fabric types and complex textures, detecting defects is very challenging. Traditional manual defect detection methods are labor-intensive and often exhibit low efficiency. In addition, these methods are constrained by human eye fatigue, which frequently results in significant missed detections.

Recently, fabric defect detection [1], [2] based on industrial vision has received increasing attention as a highly promising approach in the field of quality inspection automation [3]. Industrial cameras capture fabric images, and defect patterns are analyzed through feature extraction techniques. Early computer vision-based fabric defect detection relied on handcrafted feature methods [4], [5] to extract discriminative defect features. However, in real-world applications, fabric and defect diversity limits the effectiveness of handcrafted features in providing reliable results. Deep learning has achieved significant success in visual tasks, leading to the development of supervised fabric defect detection methods [6] such as classification, object detection, and semantic segmentation. These methods rely on large labeled datasets of defects and normal samples to train discriminative models. However, obtaining sufficient labeled defective samples in real-world scenarios is often difficult. To address this, deep unsupervised defect detection models [7], which use only normal fabric images, have been widely explored. These models learn normal patterns and detect defects by identifying significant deviations from them. Unsupervised approaches are mainly categorized into image-reconstruction-based and feature-representation-based methods. Reconstruction-based methods use generative models [8], [9], [10] to learn normal sample distributions through image reconstruction, detecting defects as deviations from the normal. However, overgeneralization can lead to accurate reconstruction of defects, causing missed detections. Feature-representation-based methods learn embeddings from normal images and compute defect scores by measuring the distance between test image embeddings and the normal distribution. However, capturing a generalized normal feature distribution is challenging without prior information.

Researchers have proposed numerous detection methods that learn representations of normal patterns from samples and identify defects by detecting significant deviations from these patterns. However, complex backgrounds often make it difficult for the features obtained by the model to distinguish between anomalies and backgrounds effectively. Therefore, this article raises a question: *can the model reasonably extract normal features from the complex background and weaken the interference of irrelevant patterns?* Fig. 1 illustrates the impact

Received 12 December 2024; revised 17 February 2025; accepted 28 February 2025. Date of publication 25 March 2025; date of current version 4 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62476171, Grant 62476172, and Grant 62206122; in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515011367 and Grant 2024A0505090003; and in part by Guangdong Provincial Key Laboratory under Grant 2023B1212060076. The Associate Editor coordinating the review process was Dr. Huan Liu. *(Corresponding author: Can Gao.)*

Jinbao Wang and Jie Zhou are with the National Engineering Laboratory for Big Data System Computing Technology, Shenzhen University, Shenzhen 518060, China (e-mail: wangjb@ieee.org; jie_jpu@163.com).

Jiayi Cheng, Can Gao, and Linlin Shen are with the College of Computer Science and Software Engineering and Shenzhen Institute of Artificial Intelligence and Robotics for Society, Shenzhen University, Shenzhen 518060, China, and also with Guangdong Key Laboratory of Intelligent Information Processing, Shenzhen 518060, China (e-mail: c13231503292@163.com; 2005gaocan@163.com; llshen@szu.edu.cn).

Digital Object Identifier 10.1109/TIM.2025.3553976

1557-9662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

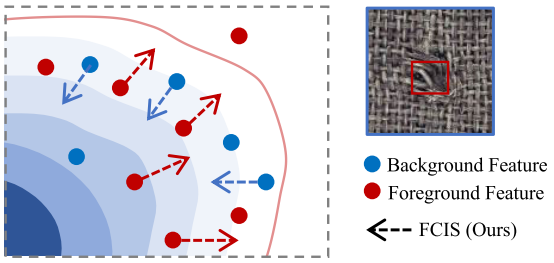


Fig. 1. Illustration of how textures affect the distribution of features. The foreground features (anomalies) are mixed with the background features (normal), making it difficult to distinguish. The proposed feature contrast interference suppression (FCIS) could enlarge the feature distance between foreground and background textures while suppressing the distribution of background features to obtain a more robust background representation.

process of complex background on the distribution of normal features. We observe that abnormal and normal features are often difficult to distinguish and mix together.

To answer the question of what features are beneficial for distinguishing anomalies, we propose a novel unsupervised framework based on teacher–student networks, which can suppress background interference and accurately segment foreground defects. However, the teacher model based on knowledge distillation is only pretrained on the ImageNet dataset [11], limiting its ability to distinguish between foreground and background features. This limitation makes the model susceptible to background interference, leading to an increased false positive rate (FPR) and reduced detection precision. To suppress fabric background interference, the teacher model can be trained to produce more robust background features and reduce their similarity to foreground features. Therefore, a background-foreground contrastive learning (BFCL) method is proposed to train the teacher model. This approach enhances the model's ability to distinguish between normal and defect features, increasing the distance between defect and normal background features in the feature space while bringing background features closer together. However, the normal feature representation learned by the student model from normal samples needs to be improved. As mentioned in [12], forcing the student model to establish normal feature representation in defect areas enhances contrast. A denoising student–teacher structure is adopted to address this, improving the student model's ability to represent the normal background, thus enabling more effective feature comparison. The anomaly score map obtained through denoising knowledge distillation lacks precise boundaries, resulting in insufficient defect localization accuracy. To address this, it can be used as an attention map alongside a segmentation module trained with pseudodefects. This approach not only preserves the generalization of unsupervised defect detection but also improves defect localization accuracy.

When facing fabrics with diverse types and textures, the existing unsupervised defect detection methods suffer from high FPRs. In this study, an unsupervised defect detection framework is proposed that reduces background interference and accurately segments foreground defects. The main contributions of this article are summarized as follows.

1) To train the teacher model for effective feature representation from complex background, a newly designed foreground–background contrastive learning method is proposed.
2) To aggregate multiscale features and effectively perform denoising knowledge distillation, a multiscale feature fusion (MFF) module is introduced.
3) To achieve high detection and segmentation precision, an attention segmentation decoder (ASD) is designed, which effectively leverages the additional supervised information in the pseudodefects.
4) Extensive experiments are conducted to verify the effectiveness of the proposed model, yielding impressive results.

The rest of this article is organized as follows. Section II provides a review of related concepts and methods. In Section III, the methodology for BFCL is described in detail. The experimental results and analysis are reported in Section IV. Finally, Section V concludes this article.

II. Related Work

A. Unsupervised Fabric Defect Detection

There are two main approaches for unsupervised fabric defect detection: image-reconstruction-based methods and feature-representation-based methods. The main concept of both the methods involves creating differences to identify anomalies in either the image space or the feature space.

For image-reconstruction-based methods, researchers use encoder networks to extract latent features and decoder networks as input to restore structured images. This type of method typically uses defect-free images as training samples, and then the differences between the input image and the reconstructed image can be used to detect defects. Recently, Schlegl et al. [13], [14] used GAN and fast-GAN for the reconstruction of fine-grained images. Paul et al. [15] used the VAE model and enhanced the image reconstruction process through structural similarity to achieve improved defect segmentation. To improve reconstruction performance, Gong et al. [16] presented an autoencoder incorporating a memory module, which encodes features as queries to extract the most pertinent memory entries. Zavrtanik et al. [17], [18] created pseudodefects on normal images to improve the model's capacity to rebuild defective regions and deployed U-Net to identify the difference between the input image and the reconstructed image. Jiang et al. [19] used masked images to enhance the reconstruction capability, introduced Transformer U-Net to inpaint the masked area, and then performed defect detection through a segmentation subnetwork. However, reconstruction-based methods may have the problem of overgeneralization, where anomalies are reconstructed into anomalies instead of normal ones during reconstruction. This means that the model cannot create a difference between the generated image and the input image.

For feature-representation-based methods, researchers make effort to extract effective features from normal fabric images and detect defects in feature space. Yi and Yoon [20] mapped the features of extracted image patches onto a hypersphere

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

to learn the distribution of normal feature representations. Roth et al. [21] used the features from these image patches for core set sampling and created a memory bank to store the normal features. During inference, when a query is presented, the model selects the nearest match from the memory bank and computes the anomaly score based on the similarity between the query and the closest match. In addition, knowledge distillation based on the teacher–student model is an important method for feature representation. Since the student network is trained on a dataset of normal samples without defects, its feature representation differs significantly from that of the pretrained and frozen teacher network. Therefore, defects can be detected by analyzing the differences in feature representations between the student and teacher models. For example, Bergmann et al. [22] integrated multiple student models to align with the teacher network's output at the patch level, which enabled fine-grained defect detection. Salehi et al. [23] enhanced both performance and inference speed by ensuring that the teacher and student models shared the same structure at the feature level. Deng and Li [24] introduced a reverse distillation framework that used a teacher model encoder and a student model decoder. This framework reversed the data flow during the knowledge distillation process, which improved the diversity of the model's defect representations. Tien et al. [25] further developed this idea by incorporating several projection layers into the teacher model, allowing it to filter out hidden defects in normal samples during training. Zhang et al. [12] used pseudoanomalies to train a denoising knowledge distillation process, ensuring a significant feature difference between the student and teacher models posttraining. However, this approach relies on a pretrained teacher model and does not consider background feature perturbations. Overall, these efforts focus on improving the similarity between student and teacher model representations on normal inputs and distancing abnormal features. However, the feature representation ability of these methods is heavily interfered by background information, resulting in large feature errors. Therefore, this article attempts to alleviate background interference, which is crucial for defect detection.

B. Supervised Fabric Defect Detection

Supervised fabric defect detection involves training a discriminative model using a substantial dataset of labeled instances. Li et al. [26] presented a denoising autoencoder based on the Fisher criterion, which uses both normal and abnormal samples for model training and identifies defect areas by applying a predetermined threshold. Liu et al. [27] developed a multistage GAN-based framework for fabric defect detection that synthesizes realistic defects within new, defect-free samples, thereby generating a substantial quantity of plausible defects to enhance the training of the classification network. Zhang et al. [28] introduced a prototype residual network that integrates multiscale prototypes and multisized self-attention mechanisms, allowing the model to learn feature residuals of varying scales and dimensions between abnormal and normal patterns, thus improving the accurate reconstruction of segmentation maps for abnormal regions. However, supervised methods depend heavily on the accuracy and quantity of data labels, which can lead to significant costs in industrial production contexts.

To further use available information (i.e., labels), semi-supervised learning can be used as a learning paradigm that compromises the unsupervised and semi-supervised learning. In semi-supervised learning methods, the main challenge is to effectively use unlabeled data to generate high-quality pseudolabels for model training. The primary strategy involves using the training model to predict labels for unlabeled data, where high-confidence predictions are referred to as pseudolabels. It is crucial to maintain consistency in the model's predictions of unlabeled data throughout the training process, ensuring that outputs remain stable despite different strengths of data augmentation. Yang et al. [29] introduced an automatic labeling module based on a similarity measurement network that uses an attention mechanism. This network generates high-quality pseudolabeled samples for training, enhancing the model's defect sensitivity. Similarly, Shao et al. [30] proposed a pixel-level semi-supervised fabric defect detection method that incorporates a multitask mean teacher approach. This method improves the model's generalization ability using a multitask supervision loss for labeled images alongside a multitask consistency loss for unlabeled images. However, the effectiveness of semi-supervised methods relies heavily on the quality of pseudolabels. The process of generating these labels can become complex, especially for unlabeled samples that exhibit diverse defects. To fully leverage abnormal information, data augmentation is a highly effective technique that can significantly enhance the model's detection performance.

C. Data Augmentation

Classic defect generation strategies, such as rotation, shearing, and mixing [31], often perform poorly in detecting fine-grained defects. Among these, simple yet effective methods known as cut-like strategies [32], [33], [34] have shown promise. Building on similar concepts, Tan et al. [35] introduced two alternative approaches for medical image detection: external patch interpolation and Poisson image interpolation. Both the methods extract patches from two images and use interpolation algorithms to synthesize the defective region in the defect-free image. However, the defects generated through these methods tend to deviate from the actual distribution of real anomalies, which limits their effectiveness in defect detection. Another strategy presented by Zavrtanik et al. [17] used 2-D Perlin noise to create more realistic anomaly masks, which are then combined with other data images to represent defects. In addition, Zavrtanik et al. [18] generated pseudodefects at the feature level using a feature representation space, ensuring that their distribution closely mirrors that of actual defects. Meanwhile, Yang et al. [36] used binarization to isolate the foreground, allowing defects to be generated solely in that area while excluding the background. However, these methods overlook the possible perturbations introduced by the background. To address this problem, our proposed model builds on the strategy from [17] for defect generation and further improves it by introducing Gaussian noise to the background

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

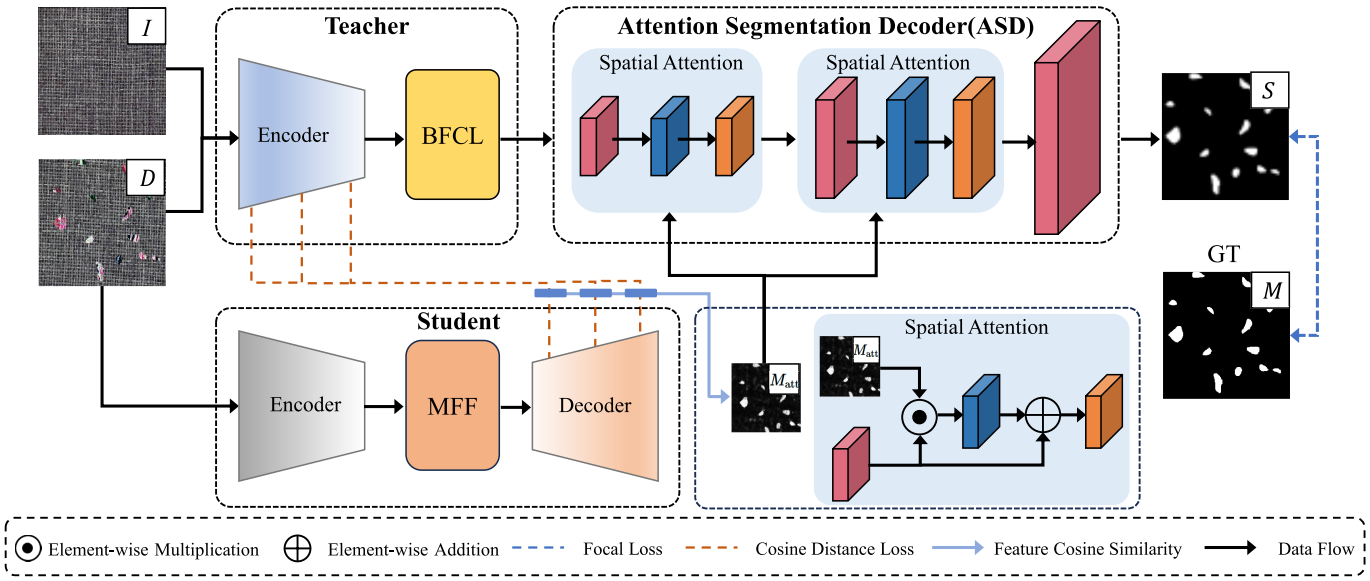


Fig. 2. Pipeline of the proposed method. First, pairs of normal images and pseudodefect images are input into the teacher model for BFCL. After BFCL training is completed, the teacher model weights are frozen for denoising knowledge distillation, which trains the student model to incorporate MFF. Finally, ASD training is performed using focal loss as the loss function.

area, thereby simulating background perturbations. As a result, the pseudodefect background generated in this way exhibits a higher level of perturbation than a typical background, providing more effective data for our proposed method BFCL.

III. Proposed Method

A. Problem Definition

Unsupervised fabric defect detection aims to identify defects using only normal samples since obtaining labeled defective samples is challenging in real-world industrial scenarios. In the training set, there are only normal samples from various fabric categories, while the test set contains both normal and abnormal samples. The task is defined as follows: given a training set $X_{\text{train}} = \{X^1_{\text{train}}, X^2_{\text{train}}, \ldots, X^n_{\text{train}}\}$ containing only normal samples and a test set $X_{\text{test}} = \{X^1_{\text{test}}, X^2_{\text{test}}, \ldots, X^n_{\text{test}}\}$. X^c_{train} and X^c_{test} represent the cth category training and test data, respectively. A corresponding model is trained for each fabric category to learn the representation of normal features. During testing, the model is used to assess whether a given sample from the test set contains any defects. If defects are detected in a sample, the entire sample is classified as defective, and the defective regions are localized.

B. Overview Framework

In this section, we will present the proposed method, which consists of three main components: BFCL, MFF module, and ASD, respectively. As shown in Fig. 2, normal and pseudodefect images are first fed into the teacher model for contrastive learning, allowing it to generate improved feature representations. After BFCL training is completed, the teacher model's weights are frozen. The pseudodefect images are then input into the student model, while the normal images are processed by the teacher model for denoising knowledge distillation training. The knowledge distillation enables the student model to learn the normal feature representation from the teacher model. Once the student model is trained, it is also frozen. The pseudodefect image is then input into both the models at the same time. An attention map is computed based on the cosine distance between the features from the two models. The output features of the teacher model, along with the attention map, are subsequently fed into the ASD to further enhance the segmentation capabilities. In Sections III-C–III-F, each module will be described in detail.

C. Defect Generation

The defect generation method described in this article adopts the DRAEM [17] strategy, as illustrated in Fig. 3. To increase the defect diversity of the image, Perlin noise is used to create irregular blocks through a binarization process. Compared with the rectangular region generation approach used in CutPaste [32], generating irregular shapes with Perlin noise more effectively simulates defects in real-world industrial scenarios. Here, we will generate a mask for pseudodefects by sampling 2-D Perlin noise P. After binarization, the defect mask is denoted as M. Next, we sample a defect texture source image A, from a dataset of abnormal source images independent of the input image N distribution. The defect texture source image A is first multiplied by the defect mask M following random data augmentation. The output is then mixed with the original image N, which has been modified by the addition of Gaussian noise. This process creates defects that slightly surpass the original data distribution, aiding in the refinement of the decision boundary during network training. Finally, the output pseudodefect image D is defined as

$$D = \overline{M} \odot g(N) + (1-\beta)(M \odot N) + \beta(M \odot A) \quad (1)$$

where $\overline{M}$ denotes the inversion of the image M, and $g(\cdot)$ represents the process of adding Gaussian noise with a mean of 0 and a variance of 0.0001. This background noise injection

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

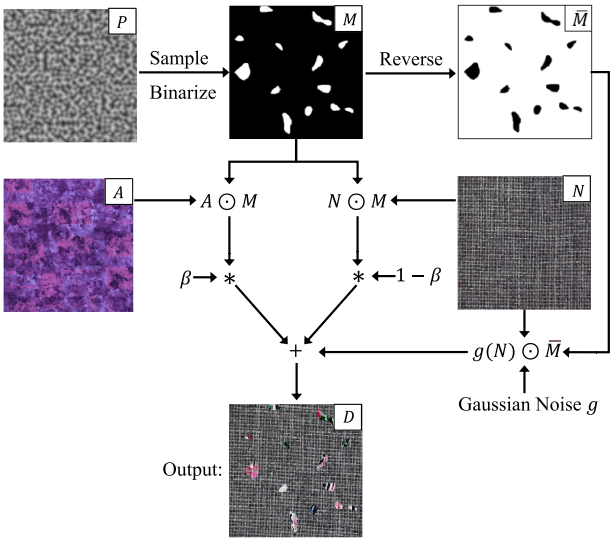


Fig. 3. Defect generation process. The binary anomaly mask M is generated from Perlin noise P. The defective regions are sampled from A according to M and placed on the normal image N to generate the anomalous image D.

strategy enhances the model's robustness to background disturbances during training. The $\odot$ is the operation of multiplying elements at corresponding positions. The transparency coefficient, β, is a random value selected from the range $[0, 0.8]$. It not only preserves the original texture characteristics of the fabric surface but also avoids the texture discontinuity issues caused by the direct covering in the CutPaste method. Each input image can randomly generate a different pseudodefect image using this strategy, with its true mask serving as supervisory information to train the model's segmentation capabilities.

D. Background–Foreground Contrastive Learning

The existing defect detection methods face challenges due to background disturbances. Detecting normal backgrounds as defects during inference can lead to misjudgments. It is well-known that deep features in a model's deep encoder are rich in structural semantics, while shallow features capture fine texture details [37]. This means that the ideal foreground and background information differ significantly in both texture and semantics, which is reflected in the feature space of each layer of the model. Based on this prior, we propose a new foreground–background contrast-based method to enhance the robustness of defect detection by eliminating complex background interference. The detailed structure of the BFCL method is illustrated in Fig. 4. In this method, the background and foreground features of an image show marked differences at both the texture and semantic levels. The greater the distance between the foreground and background features at each layer of the model, the stronger the model's ability to discriminate anomalies becomes. We state that the background features of different images should exhibit more compact differences in texture and semantics. By enhancing the distinctiveness of foreground features while minimizing the variability of background features, our method can reduce the influence of background disturbances and achieve lower misclassification.

The features of the kth block of the output for each normal sample input by the teacher model are recorded as $F_T^{k,\text{normal}}$, while the corresponding features of the kth block output for the foreground pseudodefect sample are denoted as $F_{T,\text{fore}}^k$. Both the feature sets have the same shape size, $F_{T,\text{norm}}^k, F_{T,\text{fore}}^k \in \mathbb{R}^{C_k \times H_k \times W_k}$. The contrastive learning process for the kth block of the model starts by downsampling the pseudodefect mask M to match the feature size using bilinear interpolation. This downsampled mask is then flattened to capture the foreground information and is represented as $\mathbf{P}_k \in \mathbb{R}^{H_k W_k \times 1}$. Consequently, $1-\mathbf{P}_k$ represents the background information. Next, the features associated with the foreground are computed, and similarly, the features linked to the background are also determined. These calculations are performed using the following formula:

$$\begin{aligned} \mathbf{N}_k^f &= F_{T,\text{norm}}^k \otimes \mathbf{P}_k, \quad \mathbf{N}_k^b = F_{T,\text{norm}}^k \otimes (1-\mathbf{P}_k) \\ \mathbf{A}_k^f &= F_{T,\text{fore}}^k \otimes \mathbf{P}_k, \quad \mathbf{A}_k^b = F_{T,\text{fore}}^k \otimes (1-\mathbf{P}_k) \end{aligned} \tag{2}$$

where $\otimes$ stands for matrix multiplication, and the features $F_{T,\text{norm}}^k$ and $F_{T,\text{fore}}^k$ are the first matrix transformed and then multiplied with the foreground information and background information, respectively. This operation extracts the foreground features $\mathbf{N}_k^f \in \mathbb{R}^{C_k \times 1}$ and background features $\mathbf{N}_k^b$ from the normal sample, as well as the foreground features $\mathbf{A}_k^f$ and background features $\mathbf{A}_k^b$ from the pseudodefect sample. Next, to increase the feature distance between foreground and background, the foreground features $\mathbf{N}_k^f$ and $\mathbf{A}_k^f$ are paired to form a negative example pair, with the distance between them according to the loss function $\mathcal{L}_-$. Conversely, to suppress the distribution of background features and obtain a more robust background representation, the background features $\mathbf{N}_k^b$ and $\mathbf{A}_k^b$ are paired to form a positive example pair, with the distance between them using the loss function $\mathcal{L}_+$. This contrast pair construction enables the model to develop position-awareness capabilities that align closely with downstream defect detection tasks. The total loss $\mathcal{L}_{\text{BFCL}}$ is formulated by

$$\mathcal{L}_{\text{BFCL}} = \mathcal{L}_- + \lambda \mathcal{L}_+ \tag{3}$$

where

$$\begin{aligned} \mathcal{L}_- &= -\frac{1}{K}\sum_{k=1}^{K} \log\left(1 - \text{sim}\left(\mathbf{N}_k^f, \mathbf{A}_k^f\right)\right) \\ \mathcal{L}_+ &= \frac{1}{K}\sum_{k=1}^{K} \left(1 - \text{sim}\left(\mathbf{N}_k^b, \mathbf{A}_k^b\right)\right) \end{aligned} \tag{4}$$

where λ is a hyperparameter that balances the loss function. The variable K represents the number of residual blocks learned through contrastive methods, and $\text{sim}(\cdot,\cdot)$ denotes the cosine similarity distance. This multiscale supervision strategy helps the network learn edge texture details in shallow layers while capturing semantic information in deeper layers, resulting in a feature pyramid representation that maintains texture-semantic consistency. The distance between the foreground and background features is increased for both shallow texture features and deep semantic features. While the distance

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

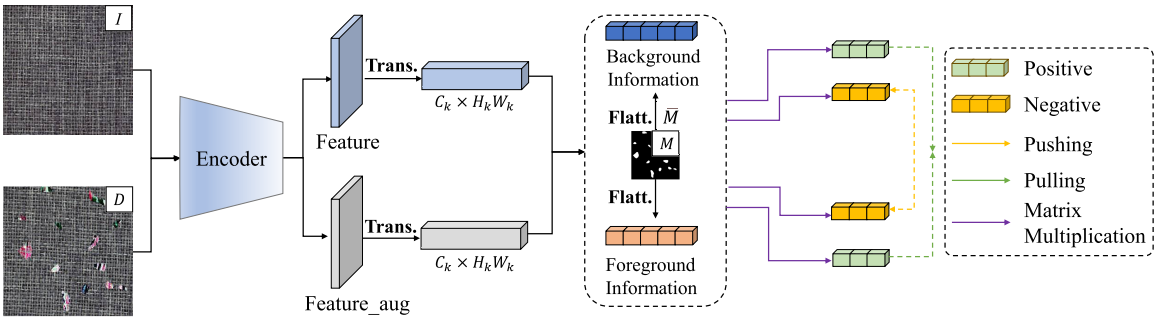


Fig. 4. Overview of BFCL. The BFCL uses feature representations from a specific encoder layer for contrastive learning, thereby constructing both positive and negative samples. Matrix transformation (Trans.) and flattening (Flatt.) are two operators used to process the data, respectively.

between the background features is reduced, resulting in a more compact representation. This reduction in distance helps minimize the perturbations caused by background features.

E. Denoising Knowledge Distillation

After contrastive learning training, the weights of the teacher model are frozen. The student model is then trained to match the output of the teacher model's normal features. Most existing knowledge distillation methods train the student model by inputting normal samples to learn the feature representation of the teacher model's normal samples. In contrast, denoising knowledge distillation uses the student model to input pseudodefect samples and re-emulate the teacher model's normal feature representation, thereby learning a denoising process. This approach enhances the model's ability to represent normal features. The feature output of the kth layer corresponding to the foreground pseudodefect sample input into the student model decoder is denoted as $F^k_{S,\text{fore}} \in \mathbb{R}^{C_k \times H_k \times W_k}$. The corresponding normal sample feature output by the teacher model is denoted as $F^k_{T,\text{norm}}$. For these two feature outputs $F^k_{T,\text{norm}}, F^k_{S,\text{fore}} \in \mathbb{R}^{C_k \times H_k \times W_k}$, calculate their vector-level cosine similarity loss and obtain a 2-D attention map $M_{\text{att}} \in \mathbb{R}^{H_k \times W_k}$

$$M_{\text{att}}(h,w) = \frac{\left(F^k_{T,\text{norm}}(h,w)\right)^{\top} \cdot F^k_{S,\text{fore}}(h,w)}{\left\|F^k_{T,\text{norm}}(h,w)\right\| \left\|F^k_{S,\text{fore}}(h,w)\right\|}. \tag{5}$$

The larger the value in M_{att}, the higher the abnormality at that location, indicating a higher probability of it being detected as a defect.

To more accurately capture the relationship between high- and low-dimensional information during knowledge transfer, the distillation loss $\mathcal{L}_{\text{KD}}$ is used, which is defined as

$$\mathcal{L}_{\text{KD}} = \sum_{k=1}^{K} \left\{ \frac{1}{H_k W_k} \sum_{h=1}^{H_k} \sum_{w=1}^{W_k} 1 - M_{\text{att}}(h,w) \right\}. \tag{6}$$

The student model functions as an autoencoder, leading to differences in the feature space between the encoder and the decoder. To enhance the effectiveness of knowledge distillation, an MFF module is used to deal with both deep and shallow encoding features, which are subsequently input into the decoder to enhance knowledge transfer. The detailed structure of the MFF module is illustrated in Fig. 5.

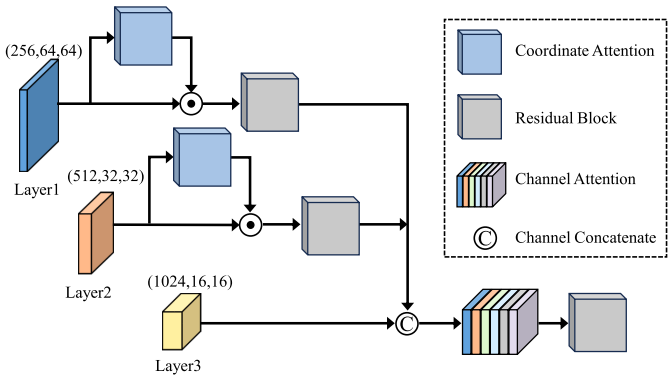


Fig. 5. Illustration of MFF.

Initially, the spatial information of the shallow features is improved using coordinate attention [38]. These enhanced features are then downsampled to align with the size of the deep features through a residual layer before being concatenated with them. In addition, channel connections are established using a channel attention mechanism [39]. Finally, the information is integrated using a residual block to produce the fused features, which are then fed into the decoder of the student model.

F. Attention Segmentation Decoder

In current knowledge distillation methods, the cosine distance of multilevel features is directly added to calculate the abnormality score for each pixel. However, if the discrimination accuracy of features at different levels is inconsistent, it can adversely affect defect detection performance. To resolve this issue, the ASD is introduced to process the features from the teacher model, ultimately producing the final segmentation map. This method effectively combines the defect detection capabilities of knowledge distillation with the strong feature representation offered by BFCL. Specifically, the ASD has three components, namely, a single pyramid scene parsing (PSP) module, three residual blocks, and an upsampling layer. The PSP module enhances the model's receptive field and improves segmentation quality, while the upsampling uses bilinear interpolation. This straightforward design allows the ASD to achieve lower memory consumption. In addition, the ASD uses a simple attention mechanism that broadcasts and multiplies spatial attention elements across residual blocks of

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DATASET INFORMATION STATISTICS

Dataset	Training set	Test set	
	Normal	Normal	Defect
MVTec AD-Carpet	280	28	89
ZJU-Leaper	37,735	18,803	6,335
DAGM	7,034	6,966	1,1054
HKU	60	15	81
AITEX	843	552	167
Total	45,952	26,364	7,726

varying sizes. The detailed process is described as

$$F_{\text{att}}^{k} = F_{\text{seg}}^{k} + F_{\text{seg}}^{k} \odot B(M_{\text{att}}) \tag{7}$$

where the term $F_{\text{att}}^{k} \in \mathbb{R}^{C_k \times H_k \times W_k}$ represents the feature output of the decoder after applying spatial attention. This feature also serves as the input to the subsequent residual block. The output feature of the current residual block is denoted as F_{seg}^{k}. The operation $B(\cdot)$ refers to the process of copying a 2-D tensor along the channel dimension. Here, $\odot$ indicates elementwise multiplication. The final 2-D anomaly score map, $S \in \mathbb{R}^{H \times W}$, is obtained by training the ASD with extra supervision derived from the pseudodefect and its corresponding mask M. The segmentation loss used is the focal loss [40], which is particularly effective in situations where there is a significant imbalance between the background and foreground in defect detection. The formula for this loss L_{seg} is provided below

$$\mathcal{L}_{\text{SEG}} = -\frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}(1-p(h,w))^{\gamma}\log(p(h,w))$$
$$p(h,w) = M(h,w)S(h,w) + (1-M(h,w))(1-S(h,w)) \tag{8}$$

where $p(h, w)$ denotes the predicted probability for each corresponding pixel in the output segmentation map, and γ serves as a hyperparameter that controls the degree of focus.

IV. EXPERIMENTS

To evaluate the effectiveness of the proposed method, a series of experiments are conducted on five benchmarks. Besides, ablation studies are performed on the MVTec AD-Carpet dataset to investigate the impact of different components on the model's performance.

A. Experimental Setup

1) Datasets and Methods: This article focuses on the task of unsupervised fabric defect detection. It leverages five defect detection datasets for comparison: MVTec AD [41], ZJU-Leaper [42], DAGM [43], HKU [44], and AITEX [45]. Fig. 6 displays some typical images from the training set. In the experiment, the proposed method is compared with the existing state-of-the-art (SOTA) method, including DRAEM [17], MKD [23], RD [24], DSR [18], DeSTSeg [12], and MemSeg [36]. Table I presents detailed information regarding both the training and test sets.

MVTec AD [41] is one of the most widely used benchmarks for anomaly detection and localization. The dataset

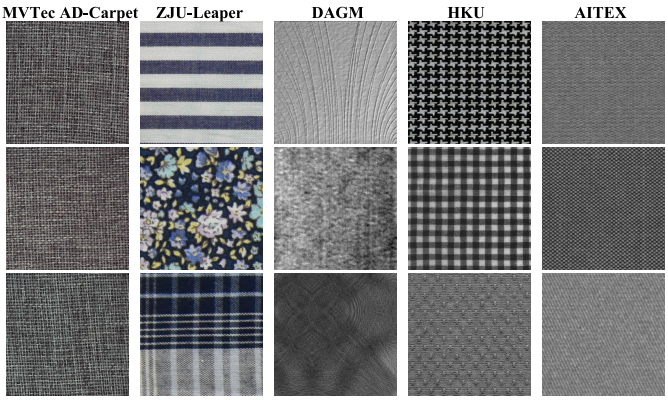


Fig. 6. Typical example images from five datasets used in the article. Each column of images comes from the same dataset.

consists of 15 categories, which include ten objects and five textures. Each category contains hundreds of normal images for training, along with a mix of abnormal and normal images for evaluation. The image sizes range from 700 × 700 to 1024 × 1024 pixels. The abnormal images in the test sets come with pixel-level binary annotations to facilitate evaluation. For our experiments, we have selected the Carpet dataset.

ZJU-Leaper [42] comprises 19 common textiles from the textile industry. The training set contains 47 423 normal samples, while the test set includes 23 704 normal samples and 23 706 defective samples. Each defective image is accompanied by a corresponding defect segmentation mask. All the images have a resolution of 512 × 512 pixels.

DAGM [43] is a well-known benchmark for surface defect detection, featuring images of various surfaces with synthetically generated defects. This dataset is organized into ten distinct classes, each representing different levels of difficulty. The DAGM consists of a total of 8050 samples across both the training and test sets. It maintains a positive-to-negative sample ratio of approximately 1:7 for each defect class.

HKU [44], provided by the University of Hong Kong, comprises three categories of fabric, each containing six types of defects: broken ends, holes, knots, multiple nets, thick stripes, and thin stripes. Each defect sample is annotated at the pixel level. The dimensions of each fabric image are 256 × 256 pixels.

AITEX [45] consists of various categories of fabric produced in industry. It includes a total of 245 images, each with a resolution of 4096 × 256 pixels, along with corresponding defect segmentation masks. In addition, the defect images are first cropped to 256 × 256, and the available fabric images are divided into training and test sets.

2) Implementation Details: In the experiment, the images in the dataset are resized to 256 × 256 pixels and normalized using the ImageNet dataset. The pretrained WRN50-2 model [46] is selected as the teacher model for BFCL training. The batch size is set to 32, consisting of 16 normal samples and 16 simulated defect samples. The maximum number of epochs is set to 20. The hyperparameters λ and γ for BFCL are set to 10 and 2, respectively, while the learning rate during training is set to 0.001. The learning rates for the denoising knowledge distillation and segmentation training stages are

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

set to 0.005 and 0.1, respectively. The training process is optimized using the Adam optimizer [47]. The batch size and the maximum number of epochs for these stages are set to 16 and 400, respectively. Our method is implemented using PyTorch 2.0.1 and CUDA 11.7, with all the experiments conducted on an NVIDIA GeForce RTX 3090 GPU.

3) Evaluation Metrics: For defect detection, the area under the ROC curve (AUC, ↑) is used to assess image-level defect detection and pixel-level defect localization. In addition, the per-region-overlap (PRO, ↑) score is introduced as another evaluation metric in the experiment (↑). PRO equally weighs the connected components of various sizes in the ground truth (GT) and calculates the overlap between the predictions and GT within a user-defined FPR, specifically at 30%. The AUC and PRO values are derived by scanning a range of thresholds and sorting the true positive rate (TPR) and FPR values accordingly. However, in surface defect detection scenarios, where only a small proportion of pixels are abnormal, the AUC does not adequately reflect localization accuracy. This limitation arises because the FPR is heavily influenced by the large number of normal pixels, causing the AUC to remain high even when there are many false positives. As noted in [48], some SOTA methods may achieve high AUC values, but the presence of background areas can still lead to numerous false positives in the segmentation map. To address this issue, the area under the precision–recall curve (AUPR/AP, ↑) has been introduced, as it is more accurate to FPR in defect localization settings.

4) Inference: During the inference stage, both the teacher and student networks process the test image. The segmentation prediction is upsampled to match the input size and is used as the anomaly score map. It is anticipated that anomalous pixels in the input image will exhibit higher values in the output. To calculate the image-level anomaly score, the top 100 values from the anomaly score map are averaged.

B. Comparison to Existing Methods

1) Image-Level Defect Detection: The performance of various methods in defect detection and localization is summarized in Table II. The proposed method achieves an average improvement of 2.38% in AUROC at the image-level across five datasets compared with other methods, showcasing enhanced robustness in defect detection. Specifically, it achieves a minimum improvement of 0.42% on the ZJU-Leaper dataset and 3.86% on the DAGM dataset. These datasets, due to their large sample sizes, present significant challenges to model robustness. On the HKU dataset, our proposed method achieves an outstanding performance of 100% across all fabric categories. In addition, on the AITEX dataset, our method outperforms the second-best method, MemSeg, by an average of 0.79%. Furthermore, the average defect detection performance across five datasets surpasses that of other methods by at least 2.38%. This consistent outperformance highlights the robustness and reliability of the proposed method, indicating its potential for effective application in real-world defect detection scenarios.

2) Pixel-Level Defect Location: The proposed method has superior defect localization performance, as shown in Table III,

TABLE II
DEFECT DETECTION EXPERIMENTAL RESULTS WITH AUROC(%) METRIC

Dataset	Category	DRAEM	MKD	RD	DSR	DeSTSeg	MemSeg	Ours
MVTec AD	Carpet	97.51	76.28	98.96	99.39	99.34	97.27	**99.42**
ZJU-Leaper	Plain	98.86	97.21	99.45	**99.93**	98.74	98.75	99.20
	Thick	99.13	99.72	**99.92**	99.76	99.78	99.90	99.72
	Thin	99.54	99.12	99.58	**99.64**	99.06	99.62	99.35
	Dot	99.65	99.86	**99.97**	99.69	99.70	97.25	99.76
	Hound	97.76	96.39	99.46	98.94	99.07	99.25	**99.50**
	Gingham	98.98	96.64	98.79	98.99	98.21	98.46	**99.22**
	Knot	98.30	97.34	98.93	99.58	**99.82**	98.91	99.21
	Twill	97.53	89.42	98.54	96.57	96.05	**98.71**	98.64
	Blue	98.00	94.68	99.13	99.18	98.07	**99.76**	99.26
	Brown	99.30	93.84	98.94	99.27	99.67	**99.70**	99.67
	Gray	83.97	86.92	98.69	95.41	98.48	99.04	**99.09**
	Red	96.01	95.16	98.87	98.15	98.22	98.59	**99.80**
	Floral1	**99.75**	82.98	98.54	99.53	99.18	99.11	99.56
	Floral2	97.09	91.06	98.42	**99.77**	98.03	98.34	98.48
	Floral3	98.26	80.80	95.31	95.59	98.13	98.01	**98.29**
	Avg.	97.48	93.41	98.84	98.67	98.68	98.01	**99.26**
DAGM	Class1	86.47	49.55	66.66	49.39	97.75	**99.91**	99.10
	Class2	87.89	82.83	98.32	93.22	97.92	**99.94**	99.54
	Class3	**99.94**	66.12	74.84	98.87	95.44	99.70	98.49
	Class4	99.19	99.08	99.62	**100.0**	99.66	96.03	**100.0**
	Class5	98.23	60.99	93.33	**99.53**	80.55	97.48	96.87
	Class6	**99.57**	90.50	78.05	55.97	98.20	82.26	99.37
	Class7	99.76	52.62	94.17	**100.0**	99.95	**100.0**	**100.0**
	Class8	**97.36**	51.42	73.90	81.31	73.96	94.99	92.43
	Class9	78.86	63.63	62.25	98.33	62.45	73.51	**96.66**
	Class10	91.26	92.22	99.83	98.65	97.35	**100.0**	99.98
	Avg.	93.85	70.90	84.10	87.53	90.32	94.38	**98.24**
HKU	box	98.46	66.92	93.86	94.61	99.20	94.62	**100.0**
	dot	**100.0**	71.00	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
	star	**100.0**	71.20	**100.0**	95.20	**100.0**	90.00	**100.0**
	Avg.	99.49	69.71	97.95	96.60	99.73	94.87	**100.0**
AITEX	Sparse	77.68	88.43	**97.89**	82.63	92.67	83.39	81.85
	Plush	73.59	70.54	74.28	**93.66**	79.31	76.33	75.07
	Stripe	87.97	**92.57**	77.54	90.90	59.92	84.12	84.05
	Dense	88.38	60.37	66.09	66.21	92.45	93.97	**100.0**
	Avg.	81.91	77.98	78.95	83.35	81.09	84.45	**85.24**
Total	**Avg.**	94.05	77.66	91.76	93.11	93.83	93.80	**96.43**

compared with other models across all the three metrics assessed. Specifically, the conventional defect localization metrics, AUROC and PRO, indicate that our method shows improvements of 0.61% and 3.73%, respectively, on average over competing models. Notably, in terms of average precision (AP), which is more sensitive to FPRs, our method exceeds the performance of other models by an average of 8.76%. When examining high-resolution images from the MVTec AD dataset, the proposed method achieves a 4.06% improvement in of AP compared with DeSTSeg, demonstrating its effectiveness in high-resolution defect detection scenarios. On the HKU dataset, given by its limited training samples and complex backgrounds, our method outperforms the second-best model, DRAEM, by an average of 3.41% in AP. For the ZJU-Leaper dataset, which consists of multicategory fabric images with a substantial sample size, our method surpasses MemSeg by an average of 5.9% in AP. In addition, on the AITEX dataset, where images are sourced from textile production, our method outperforms the second-best model, MemSeg, by an average of 5.43% in AP. Finally, on the DAGM dataset, which features small defects, our method exceeds MemSeg by an average of 6.71%. At the same time, our method outperforms competitors by 3.73% in AUPRO and 9.12% in AP. These results indicate that the proposed method effectively identifies defects of varying sizes while significantly reducing the background FPR. This enhanced robustness is attributed to the BFCL approach, which effectively suppresses

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

TABLE III

DEFECT LOCATION EXPERIMENTAL RESULTS WITH AUROC/PRO/AP(%) METRICS

Dataset	Category	DRAEM	MKD	RD	DSR	DeSTSeg	MemSeg	Ours
MVTec AD	Carpet	97.53 / 94.26 / 66.58	94.78 / 83.66 / 42.05	98.72 / 95.73 / 57.52	94.62 / 91.34 / 73.53	99.25 / 91.97 / 77.91	96.55 / 90.68 / 68.98	**99.76 / 98.22 / 81.97**
ZJU-Leaper	Plain	98.12 / 87.52 / 63.60	83.18 / 70.94 / 10.37	99.23 / 88.38 / 62.68	89.60 / 79.88 / 58.68	98.91 / 91.60 / 72.13	94.53 / 81.07 / 58.67	**99.39 / 94.32 / 78.68**
	Thick	95.76 / 79.48 / 58.03	81.80 / 59.68 / 17.57	99.10 / 69.83 / 69.80	91.93 / 83.02 / 62.69	97.18 / 90.49 / 72.84	95.39 / 85.10 / 75.76	**99.21 / 94.30 / 82.42**
	Thin	98.82 / 95.32 / 77.99	80.16 / 72.40 / 14.73	99.49 / 96.60 / 75.91	94.83 / 92.33 / 76.33	98.52 / 94.03 / 82.37	98.99 / 93.64 / 82.02	**99.63 / 98.01 / 85.87**
	Dot	96.63 / 86.79 / 58.27	83.30 / 66.68 / 16.24	98.40 / 93.19 / 54.48	87.37 / 82.62 / 58.42	96.88 / 87.66 / 65.58	91.52 / 79.41 / 56.47	**98.58 / 91.89 / 68.77**
	Hound	97.54 / 90.00 / 78.96	82.95 / 59.10 / 18.47	98.81 / **96.04** / 67.32	93.89 / 83.34 / 60.95	98.11 / 91.57 / 71.55	94.08 / 82.60 / 74.78	**99.44** / 95.42 / **85.61**
	Gingham	97.10 / 88.09 / 47.77	79.57 / 59.60 / 11.44	98.86 / 94.21 / 70.32	90.74 / 83.03 / 60.25	98.47 / 91.82 / 74.23	92.16 / 80.85 / 71.18	**99.19 / 95.62 / 76.34**
	Knot	96.00 / 81.30 / 48.40	71.11 / 49.64 / 9.70	96.30 / **89.68** / **60.27**	87.25 / 83.29 / 48.16	95.75 / 89.00 / 59.57	90.57 / 75.56 / 54.85	**96.33** / 87.90 / 58.63
	Twill	96.20 / 85.87 / 52.02	83.29 / 61.04 / 10.57	98.90 / 91.90 / 72.89	93.31 / 84.62 / 66.17	95.73 / 82.62 / 65.41	96.18 / 86.33 / 66.44	**99.07 / 93.60 / 75.86**
	Blue	96.82 / 87.86 / 63.62	79.85 / 63.24 / 16.22	98.75 / 94.91 / 72.91	94.07 / 88.97 / 66.47	97.76 / 90.95 / 77.30	96.32 / 87.32 / 73.15	**99.34 / 95.33 / 79.81**
	Brown	96.25 / 81.53 / 35.17	81.03 / 58.68 / 7.55	97.58 / 89.38 / 46.96	76.22 / 69.51 / 35.94	97.38 / 86.52 / 31.58	96.81 / 86.75 / **65.03**	**98.46 / 92.06** / 50.39
	Gray	87.72 / 67.17 / 40.29	80.83 / 56.86 / 8.81	**98.26 / 92.24** / 64.91	83.17 / 68.92 / 49.78	93.99 / 79.42 / 61.52	90.75 / 73.88 / 63.41	97.84 / 89.90 / **69.89**
	Red	89.75 / 75.78 / 34.25	79.59 / 56.78 / 8.78	96.77 / 85.54 / 43.99	84.41 / 77.02 / 48.47	93.21 / 78.09 / 47.90	92.99 / 80.51 / 56.50	**97.75 / 90.43 / 56.80**
	Floral1	97.99 / 92.83 / 64.30	76.78 / 50.40 / 9.42	96.55 / 88.97 / 63.90	93.72 / 86.24 / 55.39	93.83 / 82.56 / 60.26	96.08 / 85.46 / 67.17	**98.75 / 93.56 / 68.30**
	Floral2	97.72 / 89.45 / 59.34	71.72 / 47.20 / 6.10	98.78 / **94.07** / 64.38	96.24 / 92.35 / 69.49	98.37 / 89.82 / 75.67	94.59 / 81.31 / 69.81	**99.22** / 93.72 / **79.49**
	Floral3	95.84 / 85.02 / 58.30	63.86 / 37.79 / 5.25	95.70 / 84.58 / 58.56	93.45 / 85.24 / 57.88	96.52 / 83.47 / 74.91	93.48 / 79.95 / 68.62	**98.46 / 90.45 / 75.46**
	Avg.	95.88 / 84.93 / 56.02	78.60 / 58.00 / 11.41	98.10 / 89.97 / 63.29	90.01 / 82.69 / 58.34	96.71 / 87.31 / 66.19	94.30 / 82.65 / 66.92	**98.71 / 93.10 / 72.82**
DAGM	Class1	94.27 / 87.70 / 57.71	57.25 / 15.95 / 1.20	91.75 / 73.52 / 2.10	80.03 / 55.83 / 12.02	99.33 / 97.89 / 75.16	99.75 / **99.42 / 83.04**	**99.77** / 99.13 / 76.67
	Class2	98.33 / 96.20 / 47.36	97.72 / 89.89 / 12.04	99.57 / 98.00 / 12.10	85.98 / 69.14 / 11.50	99.88 / 99.60 / 53.45	99.22 / 96.01 / 69.17	**99.80 / 99.96 / 80.47**
	Class3	99.98 / 99.80 / 86.10	78.93 / 49.36 / 0.91	98.70 / 92.50 / 10.25	97.79 / 91.01 / 44.40	98.62 / 93.75 / 46.91	99.66 / 96.51 / 75.48	**100.0 / 98.88 / 51.04**
	Class4	99.80 / 97.10 / 38.65	96.13 / 87.49 / 7.29	99.80 / 98.23 / 13.81	99.89 / 99.23 / **83.22**	99.88 / 98.55 / 54.90	98.40 / 84.41 / 64.41	**100.0 / 99.56** / 79.68
	Class5	99.49 / **98.60 / 82.18**	72.06 / 39.53 / 12.01	**99.61** / 98.47 / 41.32	99.26 / 98.58 / 79.29	91.47 / 79.45 / 21.98	98.39 / 95.46 / 73.08	99.16 / 97.78 / 68.12
	Class6	**97.79 / 89.36** / 44.15	72.14 / 35.92 / 17.01	87.30 / 64.98 / 8.84	73.62 / 43.12 / 13.30	86.36 / 68.79 / 28.69	33.43 / 17.54 / 8.28	92.64 / 83.73 / **60.55**
	Class7	94.40 / 87.95 / 45.08	65.66 / 29.71 / 6.45	**98.82** / 89.82 / 55.53	87.41 / 75.64 / 38.51	80.23 / 65.61 / 57.21	79.63 / 64.28 / 33.86	95.77 / **94.05 / 69.28**
	Class8	**99.94 / 98.10** / 73.53	50.79 / 16.57 / 0.50	97.09 / 87.30 / 4.60	84.92 / 61.29 / 7.25	99.54 / 96.20 / 9.93	97.27 / 87.48 / 41.14	99.28 / 96.41 / 20.88
	Class9	72.73 / 31.60 / 12.53	91.73 / 76.28 / 2.10	91.86 / 73.12 / 4.13	95.51 / 86.30 / **53.26**	96.94 / 80.74 / 12.74	87.81 / 78.96 / 33.04	**99.86 / 92.38** / 46.45
	Class10	98.65 / 89.40 / 38.56	95.32 / 84.50 / 5.98	99.86 / 92.30 / 23.36	97.36 / 87.87 / 64.64	99.52 / 90.07 / 38.48	98.54 / 59.70 / **70.61**	**99.94 / 93.76** / 66.36
	Avg.	95.54 / 87.58 / 52.59	77.77 / 52.52 / 6.55	96.44 / 86.82 / 17.60	90.18 / 76.80 / 40.74	95.18 / 87.07 / 39.95	89.21 / 77.98 / 55.21	**98.62 / 95.56 / 61.95**
HKU	box	96.08 / **90.60** / 59.67	94.39 / 76.09 / 31.79	98.33 / 87.49 / 46.87	94.24 / 61.39 / **70.47**	**99.12** / 90.44 / 66.89	91.66 / 50.80 / 47.94	99.08 / 89.58 / 66.07
	dot	98.29 / **93.00** / 83.30	89.15 / 52.42 / 28.58	97.62 / 89.30 / 64.42	93.47 / 81.99 / 73.03	96.62 / 87.38 / 67.30	97.56 / 89.99 / 78.84	**98.94** / 89.66 / **83.56**
	star	99.02 / 89.90 / 73.93	95.98 / 84.09 / 23.82	99.45 / 93.29 / 64.72	93.93 / 80.65 / 54.27	99.39 / **95.38 / 78.61**	93.94 / 75.49 / 43.06	**99.55** / 94.65 / 77.49
	Avg.	97.80 / 91.17 / 72.30	93.17 / 70.87 / 28.06	98.47 / 90.03 / 58.67	93.88 / 74.68 / 65.92	98.38 / 91.07 / 70.93	94.39 / 72.09 / 56.61	**99.19 / 91.30 / 75.71**
AITEX	Sparse	78.69 / 26.19 / 6.36	46.81 / 61.87 / 2.48	97.41 / **79.44** / 34.45	83.38 / 54.67 / 12.13	**97.67** / 75.82 / 62.62	86.83 / 46.98 / 39.93	94.91 / 62.18 / **70.42**
	Plush	90.97 / 68.50 / 38.83	90.83 / 58.90 / 9.30	**92.41 / 83.17** / 11.90	91.01 / 66.42 / 58.24	88.89 / 62.96 / 24.43	83.03 / 70.53 / **60.04**	84.12 / 75.20 / 40.28
	Stripe	94.27 / **78.18** / 25.08	84.54 / 54.55 / 7.43	90.32 / 46.43 / 27.65	65.06 / 67.77 / **31.01**	86.69 / 53.58 / 24.95	87.40 / 63.93 / 18.85	**94.98** / 70.33 / 24.80
	Dense	**99.86 / 99.60 / 69.70**	90.11 / 91.96 / 13.87	98.99 / 75.77 / 47.00	69.66 / 72.11 / 16.63	91.13 / 93.59 / 17.61	97.80 / 90.17 / 21.60	99.16 / 89.14 / 36.66
	Avg.	90.95 / 68.12 / 34.99	78.07 / 66.82 / 8.27	**94.78** / 71.20 / 30.25	77.28 / 65.24 / 29.50	91.10 / 71.49 / 32.40	88.77 / 67.90 / 35.11	93.29 / **74.21 / 40.54**
Total	**Avg.**	95.54 / 85.21 / 56.50	84.48 / 66.37 / 19.27	97.30 / 86.75 / 45.47	89.19 / 78.15 / 53.61	96.12 / 85.78 / 57.48	92.64 / 78.26 / 56.57	**97.91 / 90.48 / 66.60**

TABLE IV

INFLUENCE OF DIFFERENT MODULES ON DEFECT IDENTIFICATION AND SEGMENTATION

BFCL	ASD	MFF	Image AUROC	Pixel AUROC	Pixel PRO	Pixel AP
✓			99.03	99.25	97.74	79.48
	✓		99.02	98.97	96.96	75.83
		✓	97.86	98.49	95.70	67.90
✓	✓		99.06	99.44	98.03	80.27
	✓	✓	98.97	98.22	97.31	76.78
✓		✓	98.64	99.26	97.06	75.37
✓	✓	✓	**99.42**	**99.71**	**98.22**	**81.96**

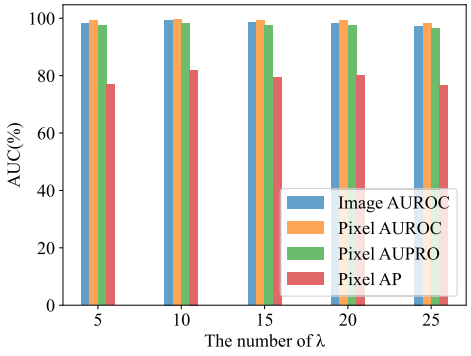


Fig. 7. Influence of different λ values on performance.

TABLE V

EFFECT OF DIFFERENT FEATURE LAYERS ON DEFECT DETECTION AND SEGMENTATION

Feature Layer	Image AUROC	Pixel AUROC	Pixel PRO	Pixel AP
1	99.08	99.19	97.57	80.28
2	99.12	99.42	97.83	80.56
3	99.04	99.13	97.29	73.90
1,2	98.80	99.04	97.26	80.52
2,3	99.20	99.34	97.99	77.88
1,3	99.40	99.40	98.12	80.63
1,2,3	99.42	99.71	98.22	81.96

background features while highlighting foreground features. As a result, the teacher model achieves superior feature representation.

C. Ablation Study

In this section, we perform ablation studies on the hyperparameters of each module and provide an analysis to validate the effectiveness of the proposed method.

1) Ablation of Modules: Table IV evaluates the effectiveness of the three proposed modules in our model. The results reveal that the BFCL module significantly enhances both defect detection and localization performance. In addition, the ASD module incorporates supervised information, which further improves the model's ability to identify and locate defects. The integration of these two modules demonstrates the synergistic benefits of combining different approaches. In contrast, the MFF module without BFCL shows only limited improvement in model performance due to interference from background features, which results in suboptimal attention maps. However, when MFF is used in conjunc-

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

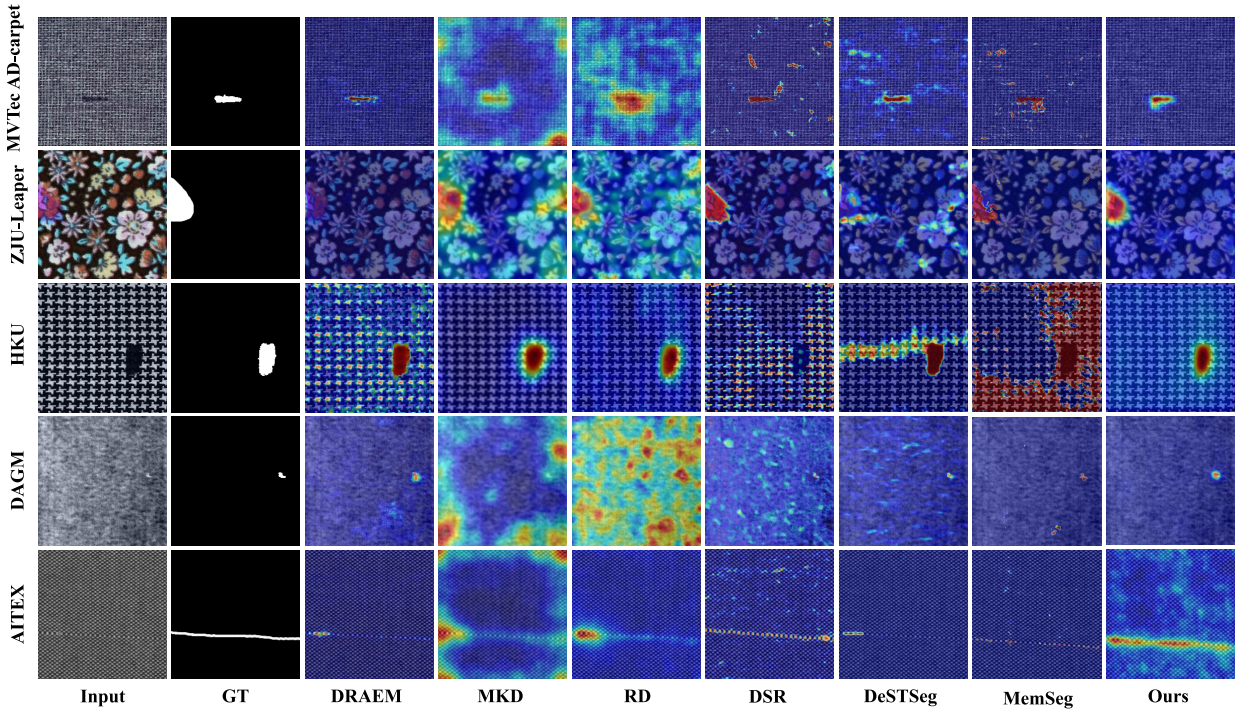


Fig. 8. Visualization of results. These images show the input, GT, and heatmaps generated by seven methods across five datasets.

tion with BFCL, the denoising knowledge distillation process becomes more effective because of the incorporation of multiscale features. This leads to a notable improvement in AP of 2.65%. Furthermore, adding the ASD module with supervised information boosts AP by 2.48%, building upon the foundations established by BFCL and MFF. This complementary relationship highlights the significance of each module in addressing various aspects of the defect detection challenge, thereby enhancing the overall effectiveness of the approach.

2) Effects of Hyperparameters: The investigation starts with analyzing the number of layers in the BFCL model. Table V summarizes how the model's detection performance varies across different feature layers trained by the teacher model. The results show that training on the shallow layers (Layers 1 and 2) leads to the most significant improvements in detection performance, while training on the deeper Layer 3 results in relatively smaller gains. This indicates that background and foreground defects in fabric share greater similarities at the texture level, whereas semantic differences become more obvious in the deeper layers. However, the coarse deep features are not sufficient for accurately identifying defect locations. Thus, a combined approach that uses both shallow texture features and deep semantic features is used can enhance detection accuracy.

Experiments are also conducted to evaluate how the hyperparameter λ affects model performance. As shown in Fig. 7, the best model performance occurs when λ is set to 10. This finding emphasizes the importance of fine-tuning hyperparameters to enhance the effectiveness of the proposed framework. Overall, the results highlight the critical roles of both feature layer selection and hyperparameter optimization in improving the robustness and accuracy of defect detection in fabric analysis.

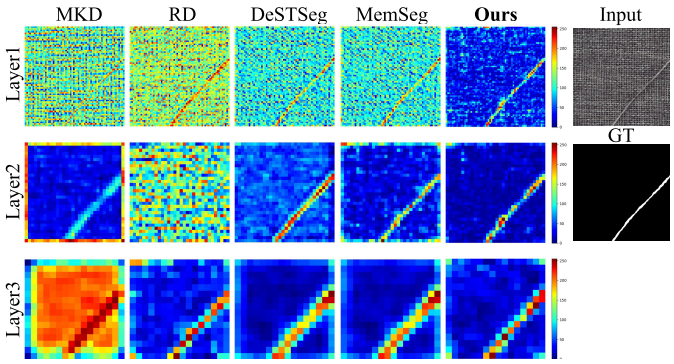


Fig. 9. Visualization of features. The images are captured from feature layers of our frozen teacher model or other feature extraction models.

D. Visualization

First, the model's segmentation output is visualized in Fig. 8. We find that our model can achieve more precise and accurate segmentation results. Besides, To demonstrate the effectiveness of BFCL, we visualize the generated feature representation, as shown in Fig. 9. It can be seen that the shallow feature representation of the model, using the BFCL method, has a notably significant impact. Compared with other models, the background features in the proposed model are more stable, and the foreground features are more easily distinguishable. These observations provide evidence for the self-interpretability of the proposed method in learning robust feature representations. Self-interpretability [49] is very important for the credibility of deep models.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

This article proposes a framework for detecting defects in fabric by addressing the challenges posed by background perturbation, which often leads to a high misclassification. To create a more robust feature of background and foreground features, we introduce a background–foreground contrast learning method. Specifically, we train an ASD using supervisory information derived from generated pseudodefects, which enhances the model's segmentation capabilities. Furthermore, we incorporate an MFF module into the knowledge distillation framework, ensuring a more comprehensive transfer of knowledge from the teacher model. Experiments conducted across several widely used benchmarks demonstrate that the proposed method effectively identifies and segments defects, achieving SOTA performance based on various evaluation metrics. The proposed method is expected to contribute to quality inspection in other industrial domains as well. Despite the above-mentioned advantages, there are still limitations that need to be addressed. First, further investigation is required to develop a unified defect detection model that can operate across different types and textures of fabrics. Second, exploring methods for detecting and identifying specific types of defects in real-world scenarios is also essential.

References

[1] A. Rasheed et al., "Fabric defect detection using computer vision techniques: A comprehensive review," *Math. Problems Eng.*, vol. 2020, pp. 1–24, Nov. 2020.

[2] A. Kumar, "Computer-vision-based fabric defect detection: A survey," *IEEE Trans. Ind. Electron.*, vol. 55, no. 1, pp. 348–363, Jan. 2008.

[3] R. Guo, H. Liu, G. Xie, and Y. Zhang, "Weld defect detection from imbalanced radiographic images based on contrast enhancement conditional generative adversarial network and transfer learning," *IEEE Sensors J.*, vol. 21, no. 9, pp. 10844–10853, May 2021.

[4] D. Chetverikov and A. Hanbury, "Finding defects in texture using regularity and local orientation," *Pattern Recognit.*, vol. 35, no. 10, pp. 2165–2180, Oct. 2002.

[5] D. Yapi, M. S. Allili, and N. Baaziz, "Automatic fabric defect detection using learning-based local textural distributions in the contourlet domain," *IEEE Trans. Autom. Sci. Eng.*, vol. 15, no. 3, pp. 1014–1026, Jul. 2018.

[6] R. Ren, T. Hung, and K. C. Tan, "A generic deep-learning-based approach for automated surface inspection," *IEEE Trans. Cybern.*, vol. 48, no. 3, pp. 929–940, Mar. 2018.

[7] D. Xu, B. Lu, and B. Huang, "A multiscale convolutional registration network for defect inspection on periodic lace surfaces," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–9, 2022.

[8] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[9] I. J. Goodfellow et al., "Generative adversarial nets," in *Proc. 27th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, vol. 27, Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, and K. Weinberger, Eds., 2014, pp. 2672–2680.

[10] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, vol. 33, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., Vancouver, BC, Canada: Curran Associates, 2020, pp. 6840–6851.

[11] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[12] X. Zhang, S. Li, X. Li, P. Huang, J. Shan, and T. Chen, "DeSTSeg: Segmentation guided denoising student-teacher for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3914–3923.

[13] T. Schlegl, P. Seeböck, S. M. Waldstein, G. Langs, and U. Schmidt-Erfurth, "F-AnoGAN: Fast unsupervised anomaly detection with generative adversarial networks," *Med. Image Anal.*, vol. 54, pp. 30–44, May 2019.

[14] T. Schlegl, P. Seebóck, S. M. Waldstein, U. Schmidt-Erfurth, and G. Langs, "Unsupervised anomaly detection with generative adversarial networks to guide marker discovery," in *Proc. Int. Conf. Inf. Process. Med. Imag.* Cham, Switzerland: Springer, May 2017, pp. 146–157.

[15] P. Bergmann, S. Löwe, M. Fauser, D. Sattlegger, and C. Steger, "Improving unsupervised defect segmentation by applying structural similarity to autoencoders," in *Proc. 14th Int. Joint Conf. Comput. Vis., Imag. Comput. Graph. Theory Appl.*, 2019, pp. 372–380.

[16] D. Gong et al., "Memorizing normality to detect anomaly: Memory-augmented deep autoencoder for unsupervised anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1705–1714.

[17] V. Zavrtanik, M. Kristan, and D. Skocaj, "DRÆM—A discriminatively trained reconstruction embedding for surface anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8310–8319.

[18] V. Zavrtanik, M. Kristan, and D. Skočaj, "DSR—A dual subspace re-projection network for surface anomaly detection," in *Proc. Comput. Vis.-ECCV*. Cham, Switzerland: Springer, Jan. 2022, pp. 539–554.

[19] J. Jiang et al., "Masked Swin transformer UNet for industrial anomaly detection," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 2200–2209, Feb. 2023.

[20] J. Yi and S. Yoon, "Patch SVDD: Patch-level SVDD for anomaly detection and segmentation," in *Proc. Asian Conf. Comput. Vis.*, vol. 12627, H. Ishikawa, C.-L. Liu, T. Pajdla, and J. Shi, Eds., Cham, Switzerland: Springer, 2021, pp. 375–390.

[21] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, and P. Gehler, "Towards total recall in industrial anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 14318–14328.

[22] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "Uninformed students: Student–teacher anomaly detection with discriminative latent embeddings," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4182–4191.

[23] M. Salehi, N. Sadjadi, S. Baselizadeh, M. H. Rohban, and H. R. Rabiee, "Multiresolution knowledge distillation for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14897–14907.

[24] H. Deng and X. Li, "Anomaly detection via reverse distillation from one-class embedding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9727–9736.

[25] T. D. Tien et al., "Revisiting reverse distillation for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 24511–24520.

[26] Y. Li, W. Zhao, and J. Pan, "Deformable patterned fabric defect detection with Fisher criterion-based deep learning," *IEEE Trans. Autom. Sci. Eng.*, vol. 14, no. 2, pp. 1256–1264, Apr. 2017.

[27] J. Liu, C. Wang, H. Su, B. Du, and D. Tao, "Multistage GAN for fabric defect detection," *IEEE Trans. Image Process.*, vol. 29, pp. 3388–3400, 2020.

[28] H. Zhang, Z. Wu, Z. Wang, Z. Chen, and Y.-G. Jiang, "Prototypical residual networks for anomaly detection and localization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 16281–16291.

[29] H. Yang, K. Song, F. Mao, and Z. Yin, "Autolabeling-enhanced active learning for cost-efficient surface defect visual classification," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–15, 2021.

[30] L. Shao, E. Zhang, Q. Ma, and M. Li, "Pixel-wise semisupervised fabric defect detection method combined with multitask mean teacher," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.

[31] H. Zhang, M. Cissé, Y. Dauphin, and D. López-Paz, "Mixup: Beyond empirical risk minimization," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2017, pp. 2866–2878.

[32] C.-L. Li, K. Sohn, J. Yoon, and T. Pfister, "CutPaste: Self-supervised learning for anomaly detection and localization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9659–9669.

[33] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6022–6031.

[34] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with cutout," 2017, *arXiv:1708.04552*.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.

[35] J. Tan, B. Hou, T. Day, J. Simpson, D. Rueckert, and B. Kainz, "Detecting outliers with Poisson image interpolation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2021, pp. 581–591.

[36] M. Yang, P. Wu, and H. Feng, "MemSeg: A semi-supervised method for image surface defect detection using differences and commonalities," *Eng. Appl. Artif. Intell.*, vol. 119, Mar. 2023, Art. no. 105835.

[37] H. Liu, B. Jiang, Y. Song, W. Huang, and C. Yang, "Rethinking image inpainting via a mutual encoder–decoder with feature equalizations," in *Proc. Comput. Vis.-ECCV*, A. Vedaldi, H. Bischof, T. Brox, and J.-M. Frahm, Eds., Cham, Switzerland: Springer, Jan. 2020, pp. 725–741.

[38] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13708–13717.

[39] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2018, pp. 7132–7141.

[40] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[41] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "MVTec AD—A comprehensive real-world dataset for unsupervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9584–9592.

[42] C. Zhang, S. Feng, X. Wang, and Y. Wang, "ZJU-Leaper: A benchmark dataset for fabric defect detection and a comparative study," *IEEE Trans. Artif. Intell.*, vol. 1, no. 3, pp. 219–232, Dec. 2020.

[43] M. Wieler, T. Hahn, and F. A. Hamprecht, "Weakly supervised learning for industrial optical inspection," in *Proc. DAGM Symp.*, vol. 6, Sep. 2007, p. 11.

[44] C. S. C. Tsang, H. Y. T. Ngan, and G. K. H. Pang, "Fabric inspection based on the Elo rating method," *Pattern Recognit.*, vol. 51, pp. 378–394, Mar. 2016.

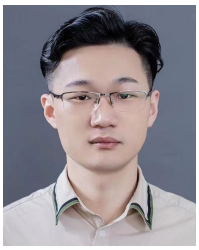
[45] J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, "A public fabric database for defect detection methods and results," *Autex Res. J.*, vol. 19, no. 4, pp. 363–374, Dec. 2019.

[46] S. Zagoruyko and N. Komodakis, "Wide residual networks," 2016, *arXiv:1605.07146*.

[47] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

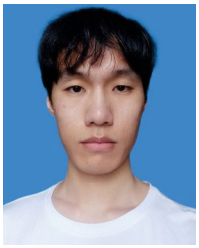
[48] X. Tao, X. Gong, X. Zhang, S. Yan, and C. Adak, "Deep learning for unsupervised anomaly localization in industrial images: A survey," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–21, 2022.

[49] R. Guo, H. Liu, G. Xie, Y. Zhang, and D. Liu, "A self-interpretable soft sensor based on deep learning and multiple attention mechanism: From data selection to sensor modeling," *IEEE Trans. Ind. Informat.*, vol. 19, no. 5, pp. 6859–6871, May 2023.



Jinbao Wang (Member, IEEE) received the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2019.

He is currently an Assistant Professor with the National Engineering Laboratory for Big Data System Computing Technology, Shenzhen University, Shenzhen, China. His research involves computer vision and machine learning, with a long-term focus on image anomaly detection, image generation, and fast retrieval.



Jiayi Cheng received the B.E. degree from Hebei University, Baoding, China, in 2023. He is currently pursuing the M.E. degree with the School of College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China.

His research interests include anomaly detection and computer vision.



Can Gao (Member, IEEE) received the Ph.D. degree from Tongji University, Shanghai, China, in 2013.

From 2010 to 2011, he was a Visiting Scholar with the University of Alberta, Edmonton, AB, Canada. From 2015 to 2018, he was a Research Associate, a Post-Doctoral Fellow, and a Research Fellow with The Hong Kong Polytechnic University, Hong Kong. He is currently an Assistant Professor with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China. He has authored or co-authored more than 60 academic articles. His current research interests include semi-supervised learning, soft computing, and anomaly detection.

Dr. Gao serves as an Associate Editor for the *Journal of Intelligent and Fuzzy Systems*.



Jie Zhou (Member, IEEE) received the Ph.D. degree from Tongji University, Shanghai, China, in 2011.

From 2010 to 2011, he was a Visiting Scholar with the Department of Electrical and Computer Engineering, University of Alberta, Edmonton, AB, Canada. From 2017 to 2018, he was an Associate Researcher at The Hong Kong Polytechnic University, Hong Kong. He is currently a Research Fellow with the National Engineering Laboratory for Big Data System Computing Technology, Shenzhen University, Shenzhen, China. His current research interests include uncertainty analysis, pattern recognition, data mining, and intelligent systems.



Linlin Shen (Senior Member, IEEE) received the Ph.D. degree from the University of Nottingham, Nottingham, U.K., in 2005.

He is currently a Pengcheng Scholar Distinguished Professor with the School of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China. His research interests include deep learning, facial recognition, analysis/synthesis, and medical image processing.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 03:23:48 UTC from IEEE Xplore. Restrictions apply.